

Urbanisation-driven nocturnal warming amplification in rapidly expanding arid cities

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Abstract

Urbanisation modifies local climate, yet the magnitude of the urban warming increment and its representation within widely used climate products remain insufficiently quantified. Here, we quantify the nocturnal urban warming increment across the United Arab Emirates during 1995–2024 by integrating long-term station observations, satellite-derived land-surface temperature, urban expansion datasets, ERA5-Land reanalysis, and NEX-GDDP-CMIP6 climate projections. Rapidly urbanising cities exhibited substantially greater nocturnal warming than locations with limited development: minimum air temperature increased by 1.31, 1.14, and 1.02 °C decade⁻¹ at Dubai, Abu Dhabi, and Sharjah, respectively, compared with 0.13–0.20 °C decade⁻¹ at stations experiencing limited urban expansion. The station-derived urban warming increment was strongly associated with surrounding built-up expansion ($\rho = 0.93$). Across the UAE, satellite observations demonstrated an exposure–response relationship, whereby each 10 percentage-point increase in built-up fraction corresponded to 0.50 °C decade⁻¹ greater nocturnal surface warming (95% CI: 0.42–0.58) relative to rural areas with comparable coastal proximity. Conversely, daytime surface-temperature trends were 0.19 °C decade⁻¹ lower than those in comparable rural areas, demonstrating a distinct diurnal thermal response to urban development. A temporal conversion analysis showed that newly urbanised areas exhibited 0.66 °C higher nocturnal temperatures during years 3–10 following conversion than similar areas that remained undeveloped, with no detectable daytime response. Rural land-surface temperature, ERA5-Land, and NEX-GDDP-CMIP6 reproduced regional background warming of approximately 0.41 °C decade⁻¹ but did not capture the additional nocturnal warming associated with urban expansion. Continued urban growth could contribute an additional 0.5–1.7 °C of nocturnal surface warming by 2050 in the fastest-growing districts. These findings demonstrate that incorporating dynamic urban characteristics is essential for improving climate assessments of heat exposure in rapidly expanding hot-desert cities.

Keywords: urban heat island; nocturnal warming; urban expansion; land-surface temperature; ERA5-Land; hot desert cities

38 1. Introduction

39 The south-eastern Arabian Peninsula is experiencing rapid warming, with regional temperatures
40 increasing at approximately twice the global mean rate and temperature extremes intensifying
41 under additional global warming (Zittis et al., 2022; Vinodhkumar et al., 2024). Along the
42 southern Gulf coast, humid heat often peaks during evening and night-time periods as maritime
43 moisture moves inland (Raymond et al., 2024). Consequently, nocturnal temperature changes
44 are particularly important for urban heat exposure because persistent warm nights limit
45 physiological recovery and increase cooling demand in cities where air-conditioning is
46 widespread (He et al., 2022).

47 Urbanisation modifies local climate through changes in surface properties, heat storage, long-
48 wave radiation exchange, and anthropogenic heat emissions, producing stronger urban heat
49 island effects during night-time periods (Oke, 1982; Manoli et al., 2019). However, urban
50 thermal responses differ substantially among climate regions. In hot desert environments,
51 irrigated vegetation, shading, and the thermal characteristics of urban materials can produce
52 daytime urban cooling relative to surrounding bare surfaces, while nocturnal warming remains
53 enhanced (Lazzarini et al., 2015). This diurnal asymmetry makes rapidly expanding desert cities
54 valuable systems for quantifying how urban development modifies local climate.

55 The United Arab Emirates (UAE) provides an exceptional setting for investigating these
56 processes because rapid urban expansion has occurred alongside pronounced regional warming.
57 Previous satellite-based studies have documented strengthening surface urban heat islands in
58 Dubai and enhanced warming in newly developed districts (Elhacham and Alpert, 2021).
59 However, the magnitude of the urban warming increment—the additional warming associated
60 with urban expansion beyond the regional climate background—remains insufficiently
61 quantified, particularly for long-term nocturnal air-temperature trends relevant to urban heat
62 assessment.

63 Climate assessments increasingly rely on reanalysis datasets and downscaled climate projections
64 to characterise present and future thermal conditions. These products provide essential
65 information for urban heat planning, yet many do not explicitly represent time-varying urban
66 expansion and may therefore underrepresent locally generated urban warming. Previous studies
67 have demonstrated that urban representation influences climate simulations from local to
68 regional scales (Zhao et al., 2021; Chakraborty and Qian, 2024; Liu et al., 2025). However,
69 whether widely used climate products capture the observed urban warming increment,
70 especially in rapidly developing arid cities, remains unresolved.

71 Here, we quantify the nocturnal urban warming increment across the UAE and evaluate its
72 representation in regional climate products. Specifically, we address two questions: (1) how
73 strongly is observed nocturnal warming associated with urban expansion, and (2) do ERA5-
74 Land and NEX-GDDP-CMIP6 reproduce this additional local warming component? We
75 integrate station observations, satellite land-surface temperature, urban expansion records,
76 reanalysis, and downscaled climate projections to separate regional background warming from
77 the additional warming associated with urban development.

2. Data and methods

2.1 Study area and study design

The United Arab Emirates (UAE) occupies the south-eastern Arabian Peninsula between 22.6–26.1°N and 51.5–56.4°E. Its arid landscape comprises extensive interior deserts, the Hajar Mountains in the northeast, and population concentrations along the Gulf coast and around inland settlements such as Al Ain (Fig. 1). Rapid land transformation against a sparsely vegetated desert background provides a distinct setting in which to evaluate the additional nocturnal warming associated with urban expansion.

The analysis integrated station air temperature, satellite land-surface temperature (LST), urban-expansion records, regional reanalysis, and downscaled climate projections. Three complementary comparisons were used: station observations against ERA5-Land and NEX-GDDP-CMIP6 at matched locations; satellite LST trends across gradients of built-up expansion relative to coast-matched rural land; and temperature trajectories before and after conversion from undeveloped to built-up land relative to matched non-converted controls. These comparisons separate regional background warming from the additional local component associated with urban expansion while maintaining the distinction between near-surface air temperature and LST.

The principal trend period was 1995–2024. Station anomalies and percentile-based indices used the 1991–2020 climatological standard normal (WMO, 2017), whereas the satellite record used 1995–2020 as its common reference period. Extreme-temperature indices followed ETCCDI definitions (Zhang et al., 2011). Future changes were evaluated relative to the 1995–2014 IPCC AR6 baseline for near-term (2021–2040), mid-century (2041–2060), and late-century (2081–2100) periods (IPCC, 2021). Urban and rural classes followed the UN Degree of Urbanisation implemented in the GHS Settlement Model grid (Dijkstra et al., 2021; Pesaresi et al., 2024). Observation–product comparisons used matched locations, periods, and baselines. Table 1 summarises the datasets and their analytical roles.

Table 1. Datasets, periods and roles. All gridded products were resampled to a common 0.009° (about 1 km) grid.

Dataset	Variable	Resolution	Period used	Role	Reference
HadISD v3.4.3.2025f	Sub-daily air and dewpoint temperature	8 stations (7 analysed)	1991–2024	Station trends, ETCCDI indices	Dunn et al. (2012)
C3S / ESA CCI LST	Monthly day and night LST	0.01°	1995–2024	Pixel-level urban signal	C3S (2023)
GHS-BUILT-S R2023A	Built-up surface	30"	1975–2020 epochs	Urban growth (Δ BF)	Pesaresi et al. (2024)
GHS-SMOD R2023A	Degree of Urbanisation classes	1 km	1995, 2000, 2020	Urban/rural definition	Dijkstra et al. (2021)
GHS-BUILT-H R2023A	Mean building height	100 m	2018	Urban form	Pesaresi et al. (2024)
Landsat C2 L2 (L5/7/8/9)	NDVI and broadband albedo	30 m → 1 km	1995–2024	Vegetation and albedo, full period	Liang (2001)
MODIS MOD13A2 / MCD43A3	NDVI; white-sky albedo	1 km / 500 m	2000–2024 / 2001–2024	Mechanism analysis	Didan (2021)
DMSP-OLS / VIIRS	Night-time light	~1 km / ~500	1996–2013 /	Urban activity, waste	Elvidge et al.

DNB		m	2014–2024	heat	(2017)
ERA5-Land	Hourly 2-m temperature → Tmin/Tmax	0.1°	1991–2024	Regional reanalysis	Muñoz-Sabater et al. (2021)
NEX-GDDP-CMIP6	Daily tasmin, tasmax; 28 models	0.25°	1991–2024; AR6 periods	Downscaled projections	Thrasher et al. (2022)
GADM 4.1	Boundaries	Vector	—	UAE and emirate masks	gadm.org

2.2 Observations and climate products

Station observations. Sub-daily 2-m air temperature and dewpoint observations were obtained from HadISD version 3.4.3.2025f, a quality-controlled subset of the Integrated Surface Database (Dunn et al., 2012; Smith et al., 2011). Eight UAE stations were initially screened after conversion to local time (UTC+4). Six stations satisfied the completeness criterion of at least 90% valid observations over the 30-year analysis period, with 29–30 valid years available for each station. Al Bateen was retained as a supplementary station due to a 2001–2009 data gap (21 valid years), whereas Al Maktoum International was excluded because observations were only available after 2011. Daily minimum and maximum temperatures (Tmin and Tmax) were aggregated to monthly anomalies and subsequently averaged to annual values. Detailed quality-control procedures and station-level processing are provided in Supplementary Methods S1.

Satellite observations. Spatially continuous evidence of urban thermal changes was derived from the Copernicus Climate Change Service (C3S) monthly land-surface temperature (LST) product based on the European Space Agency Climate Change Initiative infrared climate data record (C3S, 2023). The 0.01° dataset provides separate daytime (approximately 10:30 local time) and night-time (approximately 22:30 local time) LST observations from July 1995 to June 2025. Monthly LST values were converted to anomalies relative to each pixel’s 1995–2020 calendar-month climatology and resampled to a common 0.009° (~1 km) grid. The analysis period contained 29 usable years between 1995 and 2024 due to missing observations in 2018. Urban and rural pixels were therefore observed under consistent sensor conditions, and results were additionally evaluated across separate sensor eras. Further processing details and the assessment of MODIS as an alternative LST record are provided in Supplementary Methods S2.

Regional climate products. ERA5-Land hourly 2-m air temperature data at 0.1° resolution (Muñoz-Sabater et al., 2021) were aggregated to daily Tmin and Tmax using the same local-time definition as the station observations, followed by conversion to monthly and annual anomalies relative to the 1991–2020 climatological normal. Downscaled climate projections were obtained from NEX-GDDP-CMIP6 at 0.25° resolution (Thrasher et al., 2022). The historical dataset included 28 models with complete daily tasmin and tasmax records, with historical simulations extended through 2014 and continued using SSP2-4.5 simulations thereafter. ERA5-Land and NEX-GDDP-CMIP6 products were sampled at the nearest land grid cell for station comparisons and spatially averaged over UAE land for regional assessments. Future temperature changes were evaluated under SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5 scenarios, with the complete model inventory and exclusion criteria provided in Supplementary Table S6.

2.3 Urban expansion and morphology

Built-up surface data were obtained from GHS-BUILT-S R2023A at 30 arc-second resolution for the 1975–2020 epochs (Pesaresi et al., 2024). Built-up fraction was calculated as built-up area divided by cell area, and urban expansion was represented by the 1995–2020 change in built-up fraction (ΔBF), expressed in percentage points (pp). The expansion interval ends in

2020 because this is the final observation-supported settlement epoch. Settlement classes were obtained from GHS-SMOD, coastal distance was calculated for each land pixel, and national and emirate boundaries were obtained from GADM 4.1 (University of California, Davis, 2022).

Development-related surface characteristics comprised vegetation, albedo, and night-time light. Annual normalised difference vegetation index (NDVI) and shortwave broadband albedo were derived from Landsat Collection 2 surface reflectance for 1995–2024, including narrow-to-broadband albedo conversion (Liang, 2001). MODIS MOD13A2 NDVI (Didan, 2021) and MCD43A3 white-sky albedo (Schaaf and Wang, 2021) provided instrument-calibrated estimates for the later record. Night-time light was obtained from DMSP-OLS stable-light products and VIIRS monthly composites (Elvidge et al., 2017). DMSP-OLS changes and the 2024 VIIRS level were analysed separately because the records are not radiometrically continuous. Supplementary Methods S3 provides processing periods and endpoint definitions.

Urban morphology was characterised by mean building height (H) from GHS-BUILT-H R2023A for 2018, plan-area fraction (λ_p) from GHS-BUILT-S for 2020, and Local Climate Zones (LCZs) from the 100 m global LCZ map (Grimmond and Oke, 1999; Stewart and Oke, 2012; Demuzere et al., 2022). Height and LCZ were treated as cross-sectional modifiers of the ΔBF –warming relationship, not as temporal change variables. Aggregation rules, coverage thresholds, and alternative morphology measures are described in Supplementary Methods S3; LCZ and interaction results are reported in Supplementary Tables S3 and S4.

2.4 Urban warming estimation

For an annual temperature-anomaly series $T_{i,t}$ at station or pixel i and year t , the temporal trend was estimated using the Theil–Sen median pairwise slope (Sen, 1968; Eq. 1):

$$(1)$$

Monotonic trends were tested using the Mann–Kendall test with the Hamed–Rao variance correction for serial dependence (Hamed and Rao, 1998); ordinary least squares was retained as a sensitivity estimator. Station confidence intervals were obtained by moving-block bootstrap (Künsch, 1989). Spatial significance was evaluated by block resampling with Benjamini–Hochberg false-discovery-rate control (Benjamini and Hochberg, 1995). Supplementary Methods S4 provides the resampling specifications.

At station s , the urban warming increment U_s was estimated as the trend in the matched station-minus-ERA5-Land temperature-difference series, following the observation-minus-reanalysis framework (Kalnay and Cai, 2003; Wang et al., 2013; Eq. 2):

$$(2)$$

Here, $\beta(\cdot)$ denotes the Theil–Sen trend operator, and $T_{s,t}^{\text{obs}}$ and $T_{s,t}^{\text{ERA5-Land}}$ are the matched annual temperature anomalies. Equation (2) estimates temporal change in the local departure from the regional background rather than subtracting two independently fitted trends. Potential level shifts were assessed with the Pettitt test applied to the detrended difference series (Pettitt, 1979), followed by a shift-adjusted sensitivity estimate.

For satellite group g , the urban–rural trend contrast U_g compared the mean trend of urbanising pixels with a rural mean weighted to reproduce the urban group's distribution across coastal-distance subclasses (Cochran, 1968; Eq. 3):

$$(3)$$

Rural pixels remained in the most restrictive Degree of Urbanisation rural class in 1995 and 2020, contained less than 1% built-up surface in the pixel and its 10 km neighbourhood, and showed no detectable growth. This weighting prevents differences in proximity to the coast from being interpreted as urban effects. In Equation (3), $w_{g,c}$ is the share of group g in coastal-distance band c , and the superscripts U and R denote urbanising and rural pixels, respectively.

Warm nights were assessed using TN90p, the percentage of nights above the calendar-day 90th percentile computed from a five-day window in the 1991–2020 base period, together with tropical nights ($T_{min} \geq 20^\circ\text{C}$) and the regionally relevant count of nights with $T_{min} \geq 30^\circ\text{C}$ (Zhang et al., 2011). Each dataset used its own base-period percentiles. Coastal humidity was evaluated by removing the component of station T_{min} anomalies statistically associated with night-time dewpoint anomalies and recomputing the trend (Supplementary Methods S4).

2.5 Attribution and urban-form effects

The association between urban expansion and warming was estimated using a linear model with spatially cluster-robust inference (Cameron and Miller, 2015; Eq. 4):

$$(4)$$

In Equation (4), γ is the change in temperature trend associated with a 10 pp increase in built-up fraction; α is the intercept; x_i contains initial built-up fraction, coastal-distance band, emirate, latitude, and longitude; θ is the associated coefficient vector; and ϵ_i is the residual. Standard errors were clustered by 0.25° spatial block. NDVI change was added as a potential confounder, whereas albedo and night-time light were introduced sequentially as development-related surface covariates. Attenuation of γ was interpreted as shared statistical explanation, not causal mediation. The same specification was fitted to ERA5-Land T_{min} and T_{max} sampled at each pixel to test whether the product reproduced the local urban relationship.

A matched event study tested whether nocturnal warming followed conversion rather than preceding it, with staggered conversion cohorts represented in event time (Callaway and Sant'Anna, 2021). Treated pixels gained at least 5 pp of built-up surface from a 1995 base below 2%; controls were 61,437 never-built pixels distributed across 23 coastal-band-by-emirate strata. For treated pixel i in year t , the matched-control difference was defined as (Eq. 5):

213 (5)

214 The matched-control difference was normalised to the pixel's mean during the five pre-conversion
215 years (Callaway and Sant'Anna, 2021; Eq. 6):

216 (6)

217 In Equations (5) and (6), τ denotes years from conversion, $C(i)$ is the matched control stratum, $R_{i,t}$ is
218 the contemporaneous treated-minus-control difference and $D_{i,\tau}$ is its pre-period-normalised event-
219 time anomaly. The analysis used the 2005, 2010 and 2015 cohorts, which provide both pre- and
220 post-conversion observations. Confidence intervals and pre- and post-conversion slopes were
221 estimated within a spatial-block bootstrap. Because GHS epochs are five years apart, the conversion-
222 date threshold was tested explicitly; cohort construction, sample changes and threshold sensitivities
223 are documented in Supplementary Methods S4.

224 Modification by urban form was tested by extending the exposure–response model with the main
225 effect of H and an interaction between ΔBF and H. Results were also stratified by LCZ. Because
226 height and LCZ describe a single epoch, these analyses test whether the response to expansion
227 differs among existing urban forms; they do not estimate temporal changes in morphology.

228 2.6 Future warming scenarios

229 The estimated urban expansion–warming coefficient (γ) represents the change in warming trend
230 associated with each 10 pp increase in built-up fraction. To obtain the accumulated surface-
231 temperature response (k), γ was multiplied by the 2.9 decades spanning the 1995–2024 temperature
232 record. The resulting coefficient was applied to projected built-up expansion G_i between 2020 and
233 2050 using GHS urban-change data and the scenario framework below (Pesaresi et al., 2024; IPCC,
234 2021; Eq. 7):

235 (7)

236 The calculation assumes that the accumulated response scales with the magnitude of built-up
237 expansion rather than the rate at which expansion occurs. The predictor was averaged within 5 km of
238 each station to match the station analysis. The primary projection used the night-time LST
239 coefficient estimated from 75,528 pixels rather than the less precisely estimated six-station air-
240 temperature coefficient.

241 The 2020 GHS epoch was the projection baseline. Scenario S-high continued each cell's observed
242 2010–2020 built-up growth rate for 30 years, whereas S-low applied half that rate; both were capped
243 at the 99th percentile of observed plan-area fraction. Alternative 2005–2020 and projected 2010–
244 2025 rates were retained as sensitivity cases. The transfer coefficient was evaluated retrospectively
245 against station excess warming. Supplementary Methods S5 and Supplementary Table S5 describe

246 uncertainty propagation following standard measurement-uncertainty principles (JCGM, 2008) and
247 quantify contributions from the coefficient, growth scenario, and surface-to-air conversion.

248 **3. Results**

249 **3.1 Urban expansion and nocturnal warming**

250 UAE built-up surface increased from 298 km² in 1995 to 618 km² in 2020, while land classified
251 as urban centres expanded from 499 to 1,457 km² (Fig. 1b). Urban growth was concentrated
252 along the Dubai–Sharjah corridor, the Abu Dhabi coastal region, and around Al Ain (Fig. 1a).
253 Within 5 km of the analysed stations, built-up fraction increased by 7 percentage points (pp) at
254 Sharjah, 4 pp at Dubai, 3 pp at Abu Dhabi, and 2 pp at Al Ain, compared with 1 pp at Ras Al
255 Khaimah and Fujairah (Table 2). Urban expansion and coastal proximity were not strongly
256 aligned across the station network: Fujairah, the closest station to the coast (4.5 km),
257 experienced limited urban growth, whereas Sharjah (13.4 km from the coast) recorded the
258 largest increase in built-up fraction.

259 Urban expansion varied among emirates and across coastal-to-inland gradients. Abu Dhabi
260 accounted for the largest increase in built-up surface, adding 151 km² between 1995 and 2020
261 (47% of the national increase), followed by Dubai with 84 km² (26%) and Sharjah with 36 km²
262 (11%). Although 67% of the national increase occurred within 20 km of the coast, 18% occurred
263 more than 50 km inland, primarily around Al Ain and other interior settlements. The spatial
264 distribution of urban expansion therefore included both coastal and inland development patterns.

265 Urban expansion included both outward development and densification. The two lower growth
266 classes (2–5 and 5–10 pp) contained 2,800 of the 3,623 classified pixels in Table 3, indicating
267 that most affected pixels underwent moderate rather than extreme increases in built-up fraction.

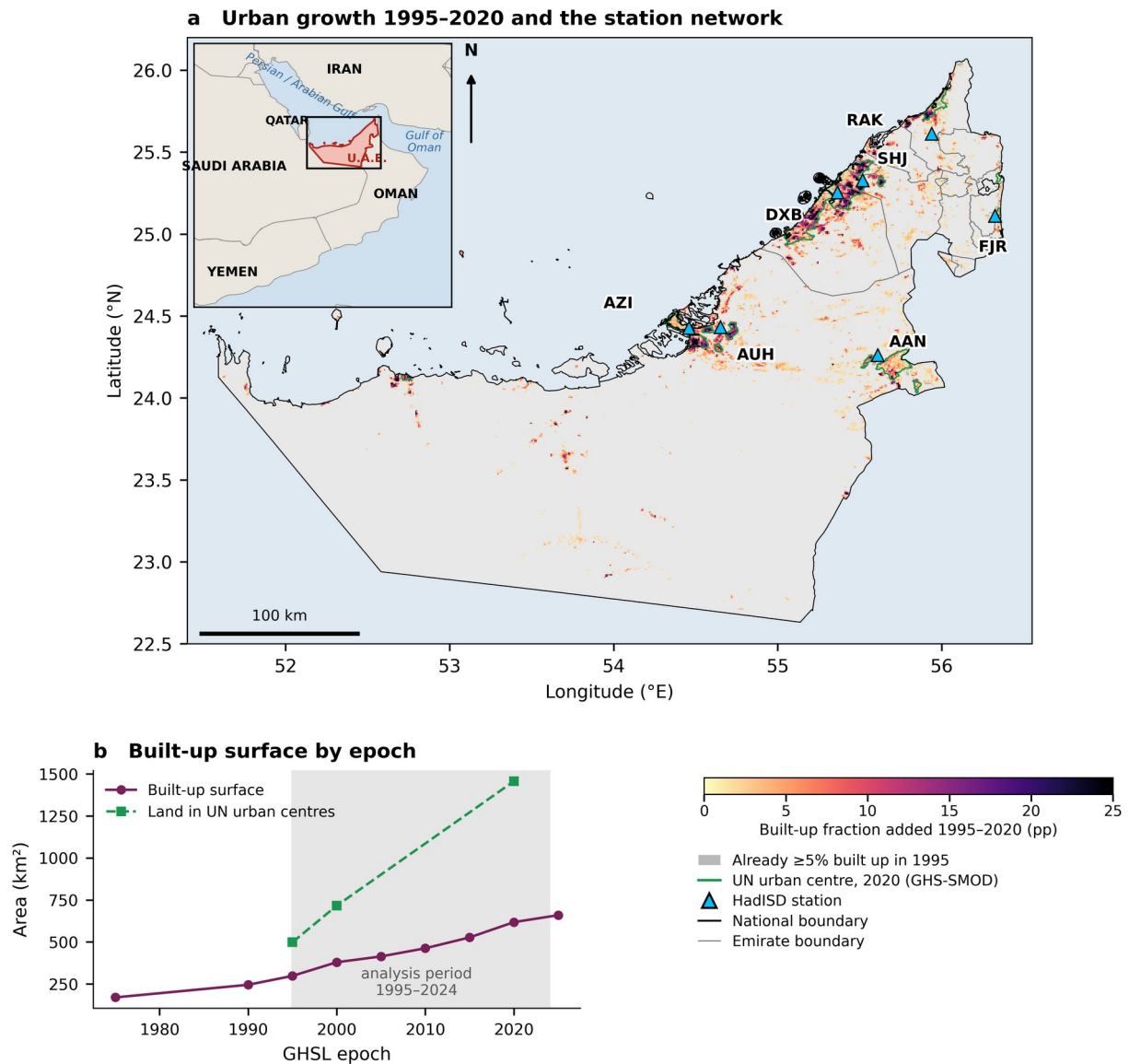


Figure 1. Urban growth and station network. (a) Increase in built-up fraction between 1995 and 2020 from GHS-BUILT-S (percentage points, shown where at least 1 pp); grey shading marks areas already at least 5% built up in 1995 and green outlines mark UN urban centres in 2020. Triangles show the HadISD stations: DXB, Dubai International; SHJ, Sharjah; AUH, Abu Dhabi International; AZI, Al Bateen; AAN, Al Ain; RAK, Ras Al Khaimah; FJR, Fujairah. (b) UAE built-up surface by GHSL epoch, and land in UN urban centres, for which the settlement grid is available for 1995, 2000 and 2020 only; shading marks the 1995–2024 analysis period. The inset locates the UAE within the Arabian Gulf region; the legend applies to both panels.

Night-time minimum temperature (T_{min}) increased faster than daytime maximum temperature (T_{max}) at all stations, although the magnitude of this difference varied spatially (Fig. 2a; Table 2). During 1995–2024, T_{min} increased by $1.31\text{ }^{\circ}\text{C decade}^{-1}$ at Dubai (95% CI: 1.04–1.60), $1.14\text{ }^{\circ}\text{C decade}^{-1}$ at Abu Dhabi (0.83–1.35), and $1.02\text{ }^{\circ}\text{C decade}^{-1}$ at Sharjah (0.72–1.29). The corresponding T_{min} – T_{max} differences were 0.80, 1.06, and $0.98\text{ }^{\circ}\text{C decade}^{-1}$, respectively. Inland Al Ain exhibited an intermediate T_{min} increase of $0.63\text{ }^{\circ}\text{C decade}^{-1}$ (0.42–0.82), whereas trends at Ras Al Khaimah ($0.20\text{ }^{\circ}\text{C decade}^{-1}$) and Fujairah ($0.13\text{ }^{\circ}\text{C decade}^{-1}$) were smaller and not statistically resolved.

ERA5-Land and NEX-GDDP-CMIP6 reproduced substantially lower Tmin trends at the rapidly urbanising stations (Fig. 2a; Table 2). ERA5-Land Tmin trends at the nearest grid cells ranged from 0.20 to 0.42 °C decade⁻¹ across stations, while NEX-GDDP-CMIP6 ensemble medians ranged from 0.33 to 0.36 °C decade⁻¹. Observed Tmin trends at Dubai, Abu Dhabi, and Sharjah exceeded all 28 NEX-GDDP-CMIP6 models at the corresponding locations, whereas Ras Al Khaimah and Fujairah were within the lower part of the ensemble distribution (21st and 11th percentiles, respectively). The station-minus-ERA5-Land urban warming increment was 1.16 °C decade⁻¹ (95% CI: 0.75–1.54) at Dubai, 0.87 °C decade⁻¹ (0.71–1.01) at Abu Dhabi, and 0.87 °C decade⁻¹ (0.54–1.25) at Sharjah, compared with 0.23 °C decade⁻¹ at Al Ain, -0.12 °C decade⁻¹ at Ras Al Khaimah, and 0.01 °C decade⁻¹ at Fujairah. Across the seven stations, the Tmin urban warming increment increased with nearby built-up expansion (Spearman $\rho = 0.93$, $p = 0.003$; Fig. 2b) but showed no relationship with distance from the coast ($\rho = -0.21$, $p = 0.64$). At Dubai, Tmin anomalies diverged from ERA5-Land after approximately 2008 and remained elevated thereafter (Fig. 2c).

The station-based urban warming increment was evaluated for sensitivity to potential discontinuities in the observational record. The Pettitt test identified a statistically significant shift only at Dubai (2019, $p = 0.04$). After removing this shift, the estimated increment at Dubai increased to 1.35 °C decade⁻¹, while estimates at Sharjah (0.81 °C decade⁻¹), Abu Dhabi (0.82 °C decade⁻¹), and Al Ain (0.26 °C decade⁻¹) remained similar (Table 2).

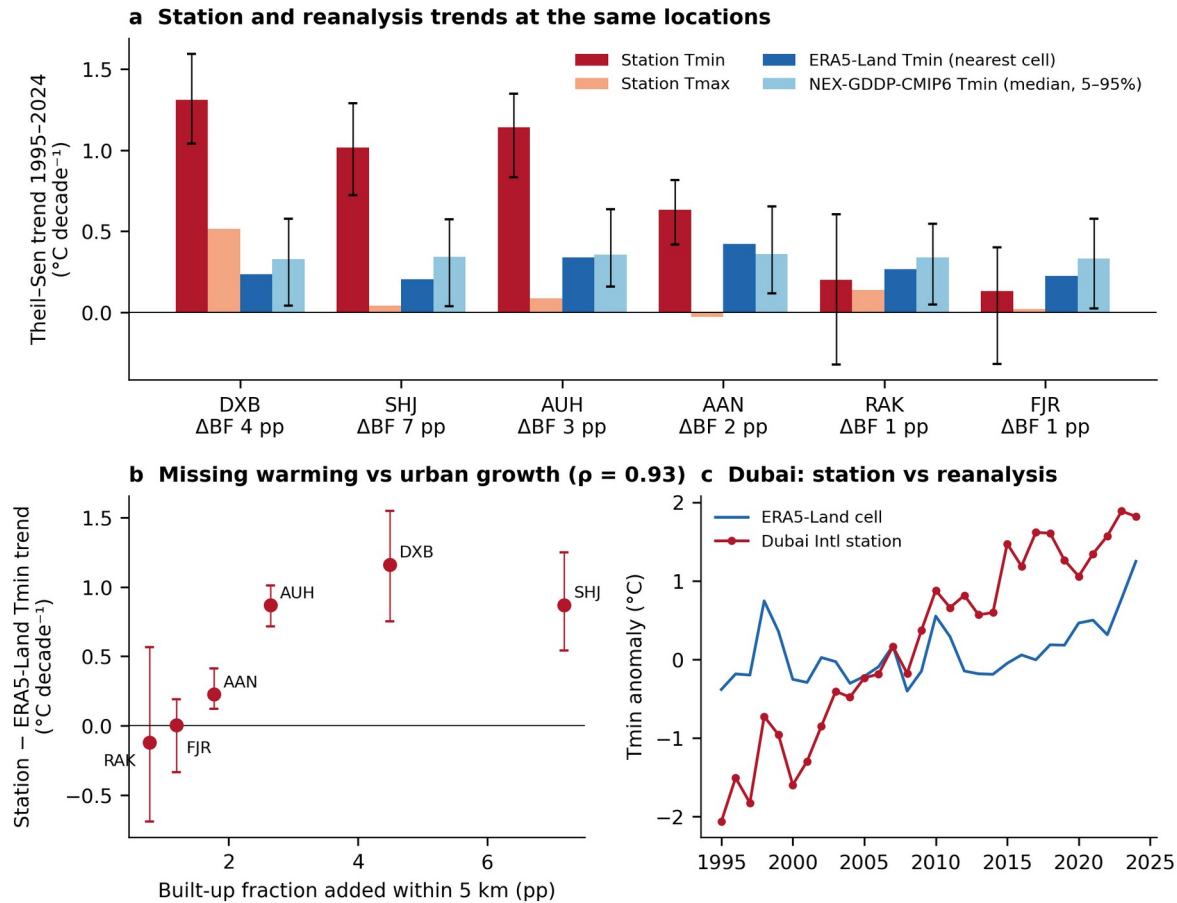


Figure 2. Station night-time warming and the regional products. (a) Theil–Sen trends for 1995–2024 in station Tmin (with 95% moving-block-bootstrap intervals) and Tmax, ERA5-Land Tmin at the nearest cell, and NEX-GDDP-CMIP6 Tmin (median and 5–95% range of 28 models) at the same locations; labels give the built-up fraction added within 5 km between 1995 and 2020 (ΔBF). (b)

309 Station-minus-ERA5-Land Tmin trend (urban warming increment) against Δ BF within 5 km, with
310 95% intervals. (c) Annual Tmin anomalies at Dubai International and the nearest ERA5-Land cell,
311 relative to 1995–2020.

Table 2. Station trends for 1995–2024 ($^{\circ}\text{C decade}^{-1}$, Theil–Sen) and comparison with regional products at the same locations. Intervals are 95% moving-block bootstrap intervals. ΔBF within 5 km is the built-up fraction added within 5 km between 1995 and 2020; NEX rank is the percentage of the 28 NEX-GDDP-CMIP6 models with a smaller T_{min} trend; the last two columns give the most likely change point in the detrended station-minus-ERA5-Land series with its Pettitt p value, and the urban warming increment recomputed after removing a shift at that year; only Dubai reaches significance.

Station	Coast (km)	ΔBF 5 km (pp)	T_{min}	T_{max}	$T_{\text{min}} - T_{\text{max}}$	ERA5-L T_{min}	NEX T_{min}	NEX rank (%)	Urban incr.	Shift year (p)	Adj. incr.
Dubai	6.7	4	1.31 (1.04–1.60)	0.51	0.80 (0.42–1.29)	0.23	0.33	100	1.16 (0.75–1.54)	2019 (0.04)	1.35
Abu Dhabi	8.8	3	1.14 (0.83–1.35)	0.09	1.06 (0.73–1.43)	0.34	0.36	100	0.87 (0.71–1.01)	1999 (0.97)	0.82
Sharjah	13.4	7	1.02 (0.72–1.29)	0.04	0.98 (0.47–1.36)	0.20	0.34	100	0.87 (0.54–1.25)	2000 (0.71)	0.81
Al Ain	98.4	2	0.63 (0.42–0.82)	-0.03	0.66 (0.24–0.91)	0.42	0.36	89	0.23 (0.12–0.41)	2005 (1.00)	0.26
Ras Al Khaimah	15.6	1	0.20 (-0.32–0.60)	0.14	0.06 (-0.42–0.65)	0.26	0.34	21	-0.12 (-0.69–0.57)	2001 (0.05)	-0.36
Fujairah	4.5	1	0.13 (-0.32–0.40)	0.02	0.11 (-0.06–0.26)	0.22	0.33	11	0.01 (-0.33–0.19)	2006 (0.95)	0.11

Al Bateen (21 of 30 years, 2001–2009 gap) is reported in Supplementary Table S7: T_{min} 0.88, urban T_{min} increment 0.49 $^{\circ}\text{C decade}^{-1}$.

3.2 Spatial and temporal attribution

Satellite LST showed enhanced nocturnal warming in areas of urban expansion (Fig. 3). Warming was concentrated along the Dubai–Sharjah corridor and its inland extensions, the Abu Dhabi metropolitan area, and parts of Al Ain. Daytime LST trends in these areas were weaker than over surrounding desert or showed cooling, whereas ERA5-Land T_{min} displayed a spatially smoother pattern without comparable urban structure. After Benjamini–Hochberg correction, 79% of land pixels had significant night-time warming trends, while no pixels had significant daytime warming trends.

Relative to coast-matched rural pixels, night-time LST warming increased systematically with the magnitude of built-up expansion (Fig. 4a; Table 3). Urbanising pixels with 2–5 pp, 5–10 pp, 10–20 pp, and ≥ 20 pp increases in built-up fraction exhibited excess night-time warming of 0.31 $^{\circ}\text{C decade}^{-1}$ (95% CI: 0.24–0.38), 0.54 $^{\circ}\text{C decade}^{-1}$ (0.41–0.65), 0.85 $^{\circ}\text{C decade}^{-1}$ (0.65–1.00), and 0.92 $^{\circ}\text{C decade}^{-1}$ (0.71–1.13), respectively. Daytime LST differences showed the opposite pattern, ranging from -0.17 to -0.54 $^{\circ}\text{C decade}^{-1}$. ERA5-Land showed no equivalent response, with T_{min} differences ranging from -0.05 to -0.07 $^{\circ}\text{C decade}^{-1}$ and T_{max} differences remaining within 0.02 $^{\circ}\text{C decade}^{-1}$. Based on the UN Degree of Urbanisation classification, urban centres warmed 0.67 $^{\circ}\text{C decade}^{-1}$ (0.50–0.81) faster than rural land at night, while urban clusters warmed by 0.47 $^{\circ}\text{C decade}^{-1}$ (0.34–0.60). After accounting for coastal band, emirate, initial built-up fraction, and spatial location, each 10 pp increase in built-up fraction was associated with a 0.50 $^{\circ}\text{C decade}^{-1}$ increase in night-time LST warming (0.42–0.58) and a -0.19 $^{\circ}\text{C decade}^{-1}$ change in daytime warming (-0.30 to -0.08). The corresponding ERA5-Land T_{min} coefficient was -0.012 $^{\circ}\text{C decade}^{-1}$ (-0.023 to 0.000). Areas that were already built-up in 1995 but experienced limited subsequent expansion also showed higher night-time warming relative to rural land (0.37 $^{\circ}\text{C decade}^{-1}$; 0.25–0.50).

The urban–rural difference in night-time LST widened progressively over the analysis period (Fig. S1). The rural background warming signal was consistent across datasets. Rural night-time

LST increased at $0.41\text{ }^{\circ}\text{C decade}^{-1}$, matching the ERA5-Land Tmin trend of $0.41\text{ }^{\circ}\text{C decade}^{-1}$ at the same pixels and the UAE-wide ERA5-Land trend ($0.41\text{ }^{\circ}\text{C decade}^{-1}$). The NEX-GDDP-CMIP6 ensemble median also indicated $0.41\text{ }^{\circ}\text{C decade}^{-1}$ warming (5–95% range: $0.18\text{--}0.63$). Thus, the central estimates of the products were consistent with rural warming, whereas the additional relationship between local urban expansion and nocturnal warming was not reproduced.

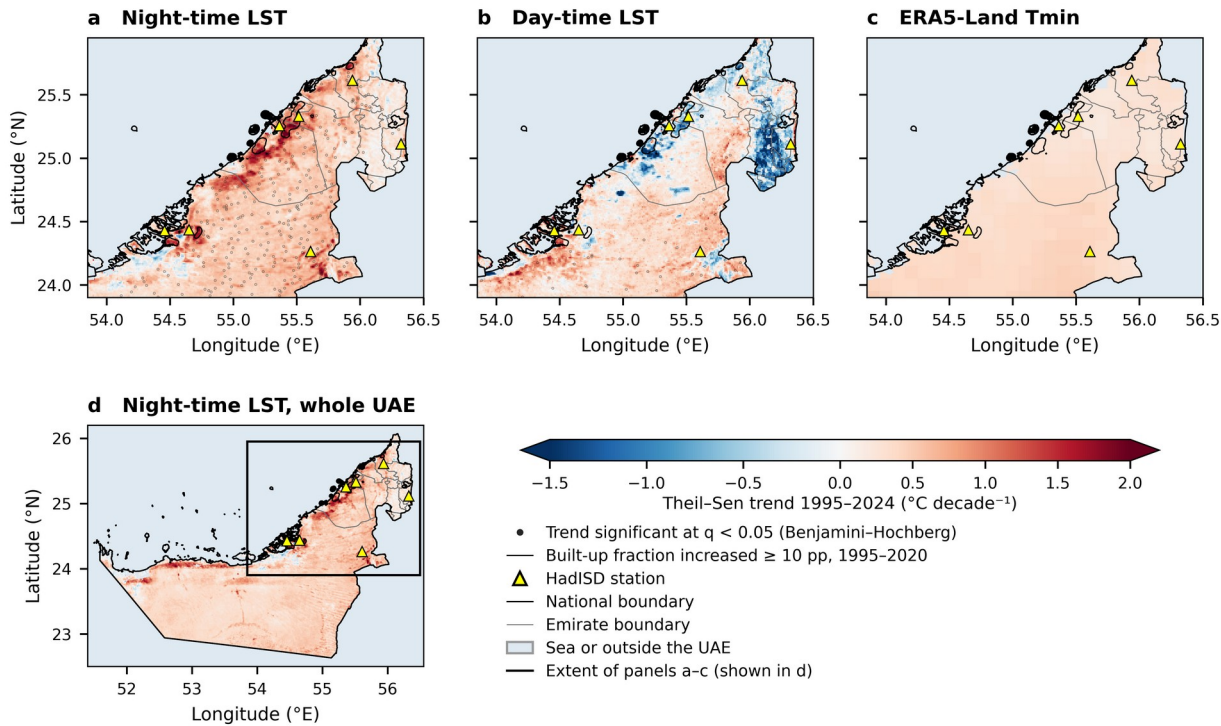


Figure 3. Theil–Sen trends for 1995–2024 over the northern UAE. (a) Night-time and (b) daytime satellite land-surface temperature (LST), with stippling where the Benjamini–Hochberg false-discovery rate is below 5%. (c) ERA5-Land Tmin on the same grid. (d) Night-time LST over the whole UAE, with the box marking the extent of panels a–c. Black contours enclose areas where built-up fraction increased by at least 10 percentage points between 1995 and 2020; triangles mark stations, and the shared colour bar and symbols are given in the legend.

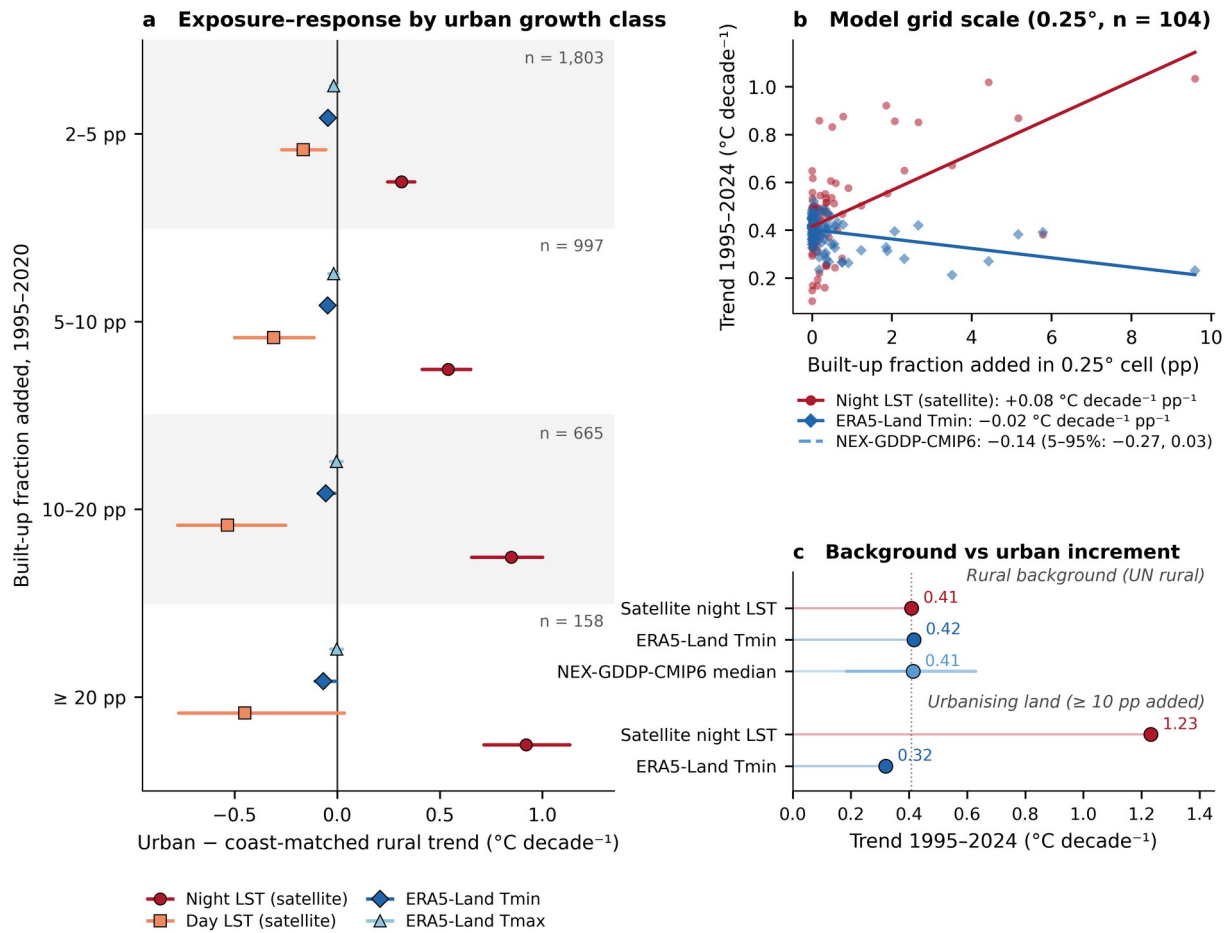


Figure 4. Exposure–response of warming trends to urban growth. (a) Trend difference between urbanising and coast-matched rural pixels by class of added built-up fraction, for night-time and daytime LST and for ERA5-Land Tmin and Tmax at the same pixels; error bars are 95% spatial block-bootstrap intervals and n is the number of urban pixels. (b) Night-time LST and ERA5-Land Tmin trends in the 104 NEX-GDDP-CMIP6 0.25° cells over UAE land against cell-mean added built-up fraction, with least-squares fits; the legend gives each slope and the NEX-GDDP-CMIP6 ensemble slope. (c) The same trends over rural land and over urbanising land (≥ 10 pp added), with the NEX-GDDP-CMIP6 ensemble median and 5–95% range; the dotted line marks the $0.41^{\circ}\text{C decade}^{-1}$ rural background.

Table 3. Urbanising minus coast-matched rural trends for 1995–2024 ($^{\circ}\text{C decade}^{-1}$). Rural land follows the UN Degree of Urbanisation. Intervals are 95% spatial block-bootstrap intervals over 0.25° blocks.

Class	Pixels	Night LST	Day LST	ERA5-Land Tmin	ERA5-Land Tmax
2–5 pp	1803	0.31 (0.24 to 0.38)	-0.17 (-0.27 to -0.06)	-0.05 (-0.07 to -0.02)	-0.02 (-0.03 to -0.00)
5–10 pp	997	0.54 (0.41 to 0.65)	-0.31 (-0.50 to -0.11)	-0.05 (-0.08 to -0.01)	-0.02 (-0.04 to 0.00)
10–20 pp	665	0.85 (0.65 to 1.00)	-0.54 (-0.78 to -0.25)	-0.06 (-0.09 to -0.01)	-0.00 (-0.03 to 0.02)
≥ 20 pp	158	0.92 (0.71 to 1.13)	-0.45 (-0.77 to 0.03)	-0.07 (-0.11 to -0.00)	-0.00 (-0.03 to 0.02)
Built up by 1995, little change	535	0.37 (0.25 to 0.50)	-0.46 (-0.66 to -0.20)	-0.06 (-0.09 to -0.03)	-0.07 (-0.09 to -0.05)
UN urban centre (2020)	1395	0.67 (0.50 to 0.81)	-0.49 (-0.72 to -0.23)	-0.06 (-0.10 to -0.02)	-0.04 (-0.07 to -0.00)
UN urban cluster (2020)	1233	0.47 (0.34 to 0.60)	-0.31 (-0.51 to -0.13)	-0.05 (-0.08 to -0.02)	-0.02 (-0.05 to 0.00)

Aligning pixels by conversion year provided an independent temporal test (Fig. 5). Among treated pixels with usable pre-conversion records (432 at night and 433 by day across 51 blocks), night-time differences from matched controls were stable before conversion and increased afterwards. The pre-conversion slope was $0.022\text{ }^{\circ}\text{C yr}^{-1}$ (95% CI: -0.001 to 0.042 , $p = 0.065$), compared with $0.096\text{ }^{\circ}\text{C yr}^{-1}$ after conversion (0.061 – 0.129 , $p = 0.0005$). During years 3–10 after conversion, treated pixels were $0.66\text{ }^{\circ}\text{C}$ warmer at night than matched controls (0.44 – 0.88 , $p = 0.0005$). No daytime change was resolved: the pre- and post-conversion slopes were $-0.032\text{ }^{\circ}\text{C yr}^{-1}$ ($p = 0.25$) and $0.049\text{ }^{\circ}\text{C yr}^{-1}$ ($p = 0.19$), and the post-conversion level difference was $0.13\text{ }^{\circ}\text{C}$ (-0.19 to 0.38 , $p = 0.41$).

The sensitivity of the conversion analysis to the assigned conversion threshold was evaluated by varying the built-up increase threshold from 1 to 5 pp (Fig. 5c). Increasing the threshold modified the estimated pre-conversion slope from $0.022\text{ }^{\circ}\text{C yr}^{-1}$ ($p = 0.069$) to $0.073\text{ }^{\circ}\text{C yr}^{-1}$ ($p = 0.001$), whereas post-conversion slopes remained within 0.061 – $0.096\text{ }^{\circ}\text{C yr}^{-1}$ and the 3–10-year post-conversion warming level remained between 0.66 and $0.84\text{ }^{\circ}\text{C}$. The treated sample size varied from 432 to 458 pixels across thresholds, indicating that the timing estimates were sensitive to the conversion definition, while the magnitude of the post-conversion warming response remained relatively stable.

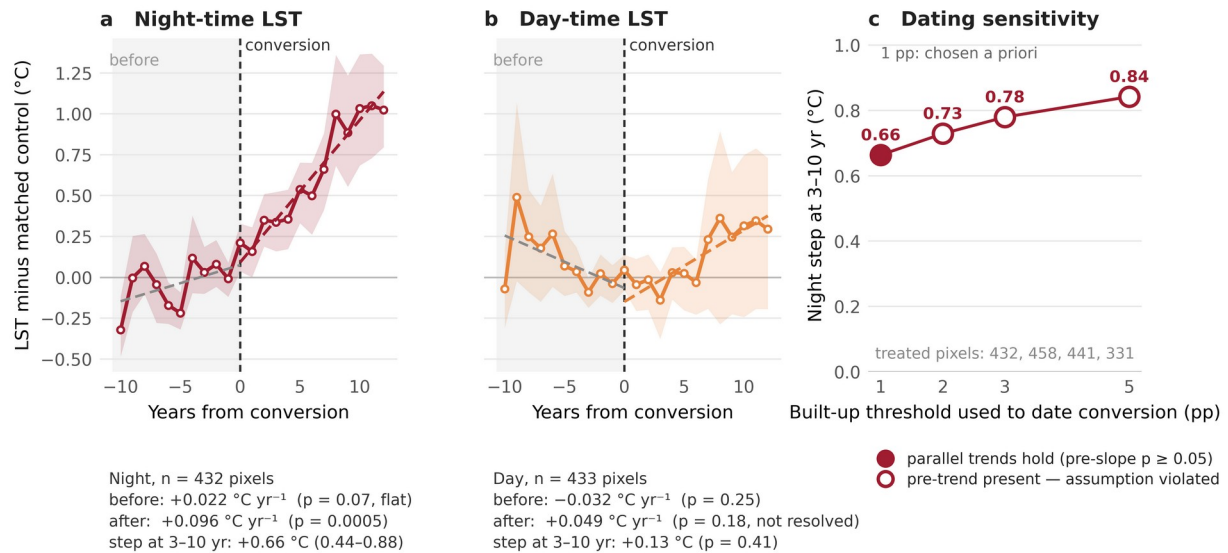


Figure 5. Night-time land-surface temperature before and after built-up conversion. Each treated pixel is differenced against the mean of never-built control pixels in the same coastal-distance band and emirate, and normalised to its own mean over the five years before conversion; shading is the 95% block-bootstrap interval over 0.25° blocks. (a) Night-time and (b) day-time difference from matched controls against years from conversion, for the treated pixels of the 2005, 2010 and 2015 cohorts (432 at night, 433 by day); the dashed line marks the conversion date and grey shading the pre-conversion window. (c) Sensitivity to the built-up threshold used to date conversion: the mean night-time level three to ten years after conversion at each threshold. Filled markers denote datings for which the pre-conversion slope is not resolved ($p \geq 0.05$), so that the parallel-trends assumption holds; open markers denote datings with a resolved pre-trend, for which the estimate is not interpretable. Only the 1 pp dating, chosen a priori, satisfies the assumption. Slopes, levels and their p values are estimated inside each bootstrap replicate.

The nocturnal warming increment remained positive across satellite sensor eras, alternative analysis periods, trend estimators, coastal exclusions, and rural-reference definitions (Supplementary Results R1; Supplementary Table S8). Although night-time slopes differed significantly among sensor eras,

a residual sensor contribution cannot be excluded; no corresponding era-related heterogeneity was detected for daytime trends.

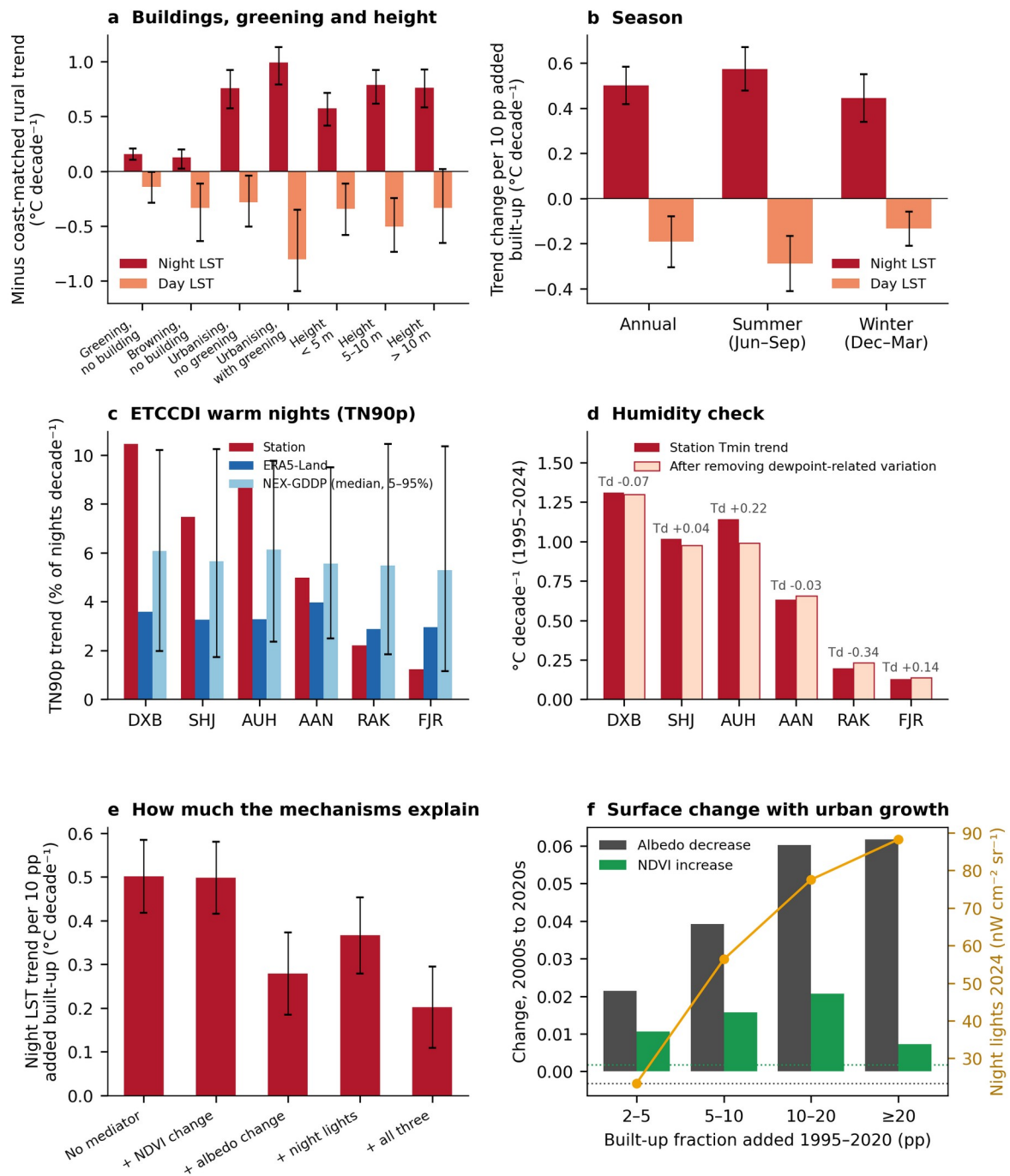
3.3 Mechanisms and urban-form effects

Greening alone was not associated with the magnitude of the nocturnal warming increment observed in urbanising areas (Fig. 6a). Land that experienced greening without built-up expansion ($\Delta\text{NDVI} \geq 0.05$, $\Delta\text{BF} < 1$ pp) warmed by $0.16\text{ }^{\circ}\text{C decade}^{-1}$ (0.10–0.21) relative to rural desert at night and exhibited daytime cooling ($-0.14\text{ }^{\circ}\text{C decade}^{-1}$; -0.29 to -0.01). Land that lost vegetation without built-up expansion warmed by $0.12\text{ }^{\circ}\text{C decade}^{-1}$ (0.02–0.20) at night. In contrast, urbanising land ($\Delta\text{BF} \geq 10$ pp) warmed by $0.76\text{ }^{\circ}\text{C decade}^{-1}$ (0.58–0.93) faster than rural areas where NDVI did not increase and by $0.99\text{ }^{\circ}\text{C decade}^{-1}$ (0.79–1.13) where NDVI increased, with corresponding daytime trends of 0.28 and $0.80\text{ }^{\circ}\text{C decade}^{-1}$ lower than rural land. Adding NDVI change and initial NDVI to the regression did not materially alter the built-up coefficient, which remained $0.49\text{ }^{\circ}\text{C decade}^{-1}$ per 10 pp (0.41–0.58).

Two measured surface characteristics substantially reduced the adjusted built-up coefficient (Fig. 6e,f). Relative to rural land, urbanising pixels ($\Delta\text{BF} \geq 10$ pp) exhibited reduced albedo (-0.061 compared with $+0.003$) and substantially higher night-time light emissions (median approximately 80 compared with $0.5\text{ nW cm}^{-2}\text{ sr}^{-1}$), while NDVI increased only slightly ($+0.018$ compared with $+0.002$). These changes scaled with the magnitude of urban expansion: mean albedo declined by 0.021 in pixels gaining 2–5 pp and by 0.062 in pixels gaining ≥ 20 pp of built-up fraction. Incorporating albedo change into the regression reduced the built-up coefficient from 0.50 to $0.28\text{ }^{\circ}\text{C decade}^{-1}$ per 10 pp, while adding night-time light reduced it to 0.37 and including all three variables reduced it to 0.20. The full-period Landsat and DMSP estimates produced a smaller but consistent reduction (0.50 to 0.35). Adding NDVI change alone did not alter the coefficient (0.498 to 0.498), indicating that the reduction was associated primarily with albedo and night-time light changes rather than vegetation change. Including surface darkening and night-time light reduced the adjusted built-up coefficient by 60% using instrument-calibrated MODIS and VIIRS records and by 31% using the noisier full-period Landsat and DMSP records. Because built-up fraction, albedo, and light emissions co-vary during urban development, these reductions represent shared explanatory contributions rather than independent component partitions.

Building height did not amplify the nocturnal warming increment; instead, the marginal warming response associated with additional built-up expansion declined with increasing building height. Night-time excess warming trends were $0.57\text{ }^{\circ}\text{C decade}^{-1}$ in urbanising pixels with mean building heights below 5 m, $0.79\text{ }^{\circ}\text{C decade}^{-1}$ at 5–10 m, and $0.76\text{ }^{\circ}\text{C decade}^{-1}$ above 10 m, with corresponding mean increases in built-up fraction of 8, 13, and 16 pp. The urban nocturnal warming increment was larger during summer, reaching $0.57\text{ }^{\circ}\text{C decade}^{-1}$ per 10 pp (0.48–0.67) during June–September compared with $0.45\text{ }^{\circ}\text{C decade}^{-1}$ (0.34–0.55) during December–March (Fig. 6b).

Humidity-related changes did not explain the station-level warming differences (Fig. 6d). Night-time dewpoint trends at urbanising stations were small (-0.07 to $+0.22\text{ }^{\circ}\text{C decade}^{-1}$) and showed no relationship with urban growth. Removing the component statistically associated with dewpoint changes reduced the station T_{min} trends only slightly: from 1.31 to $1.30\text{ }^{\circ}\text{C decade}^{-1}$ at Dubai, from 1.02 to $0.98\text{ }^{\circ}\text{C decade}^{-1}$ at Sharjah, and from 1.14 to $0.99\text{ }^{\circ}\text{C decade}^{-1}$ at Abu Dhabi. The largest adjustment occurred at Abu Dhabi, where the reduction was 13%, while the adjusted trend remained more than twice the ERA5-Land trend.



449

450 **Figure 6.** Mechanisms, alternative explanations and warm nights. (a) Night-time and daytime LST trend minus the
 451 coast-matched rural trend for land that changed without building (greening, $\Delta\text{NDVI} \geq 0.05$; browning, $\Delta\text{NDVI} \leq$
 452 -0.02), for urbanising land (added built-up fraction ≥ 10 pp) without and with greening, and for urbanising pixels
 453 (≥ 5 pp) by mean building height (< 5 m, 5–10 m, > 10 m). (b) Regression coefficient per 10 pp of added built-up
 454 fraction for annual, summer and winter LST trends. (c) Trends in the ETCCDI warm-night index TN90p (base
 455 period 1991–2020) at stations, the nearest ERA5-Land cell and NEX-GDDP-CMIP6 (median and 5–95% of 28
 456 models). (d) Station Tmin trends before and after removing the component related to night-time dewpoint; labels
 457 give the dewpoint trend ($^{\circ}\text{C decade}^{-1}$). (e) The built-up coefficient for night-time LST when satellite-measured
 458 mechanisms are added to the regression. (f) Decrease in albedo and change in NDVI between the early 2000s and
 459 the early 2020s, and night-time light emission in 2024, by class of added built-up fraction; dotted lines mark the
 460 rural values. Error bars, where shown, are 95% intervals; panels d and f show point estimates only.

Building height modified the response to built-up expansion rather than amplifying it (Table S4; Fig. S3a). Among urbanising pixels, the night-time exposure–response was $0.44\text{ }^{\circ}\text{C decade}^{-1}$ per 10 pp (95% CI: 0.34–0.55) where mean building height was below 5 m, $0.34\text{ }^{\circ}\text{C decade}^{-1}$ (0.25–0.43) at 5–10 m, and $0.14\text{ }^{\circ}\text{C decade}^{-1}$ (–0.01 to 0.29) above 10 m, with the highest-height category not distinguishable from zero. The interaction between added built-up fraction and building height was correspondingly negative, with a coefficient of $-0.36\text{ }^{\circ}\text{C decade}^{-1}$ per 10 pp per 10 m (95% CI: –0.48 to –0.25).

The positive height coefficient in the same model ($0.22\text{ }^{\circ}\text{C decade}^{-1}$ per 10 m; 95% CI: 0.13–0.30) represents the height effect at zero built-up expansion, corresponding to areas already developed in 1995, rather than indicating a general increase in warming with building height. Accounting for the interaction term, the net height effect crossed zero at approximately 6 pp of added built-up fraction and reached $-0.03\text{ }^{\circ}\text{C decade}^{-1}$ per 10 m at the mean expansion level of urbanising pixels (6.8 pp).

The negative height interaction was not explained solely by saturation in previously developed areas. An interaction between initial built-up fraction and subsequent expansion yielded a coefficient of -0.28 (95% CI: –0.39 to –0.18), while restriction to pixels with less than 10% initial built-up fraction yielded -0.32 (–0.44 to –0.19). The bulk-height index did not resolve the relationship ($0.11\text{ }^{\circ}\text{C decade}^{-1}$ per 10 m; –0.07 to 0.28).

Daytime LST showed no height effect (main effect $p = 0.81$; interaction $p = 0.30$). The ERA5-Land interaction was statistically significant ($0.03\text{ }^{\circ}\text{C decade}^{-1}$ per 10 pp per 10 m; 95% CI: 0.00–0.05; $p = 0.04$) but was an order of magnitude smaller than the corresponding night-time LST interaction and exhibited the opposite sign.

LCZ stratification showed that the urban warming increment per unit of added built-up area was greatest in sparsely built environments and smaller in the resolved low-rise and industrial classes. Differences in growth among LCZ classes and limited representation of high-rise classes at 1 km resolution prevented a complete classification-based comparison (Supplementary Results R2; Supplementary Table S3).

3.4 Warm-night indices and product representation

Warm-night indices further distinguished rapidly urbanising stations from the regional products (Fig. 6c). TN90p increased by 7.5–10.5 percentage points per decade at Dubai, Abu Dhabi, and Sharjah, compared with 3.3–3.6 percentage points per decade in ERA5-Land. Tropical nights increased by 17.8–22.2 nights decade^{-1} at these stations, compared with 2.7–4.3 nights decade^{-1} at stations with limited growth. At Dubai, nights with $T_{\min} \geq 30\text{ }^{\circ}\text{C}$ increased from 55 yr^{-1} in 1995–1999 to 126 yr^{-1} in 2020–2024. Station-level values are reported in Supplementary Results R3 and Supplementary Table S1.

The discrepancy between observations and climate products was not explained solely by spatial resolution. After aggregation to the 104 NEX-GDDP-CMIP6 0.25° grid cells covering UAE land, cells with greater built-up expansion continued to exhibit higher night-time warming rates, with an increase of $0.76\text{ }^{\circ}\text{C decade}^{-1}$ per 10 pp increase in cell-mean ΔBF ($p < 10^{-9}$; Fig. 4b). In contrast, ERA5-Land $T_{\min}$ trends decreased slightly with increasing ΔBF ($-0.20\text{ }^{\circ}\text{C decade}^{-1}$ per 10 pp), as did the NEX-GDDP-CMIP6 ensemble response (median: $-0.14\text{ }^{\circ}\text{C decade}^{-1}$ per 10 pp; 5–95% range: –0.27 to 0.03). The strongest urbanised cells are located primarily along the coast, where regional background warming is slightly weaker, contributing to this negative

relationship. However, the aggregated climate products did not reproduce the additional nocturnal warming associated with urban expansion.

Outside urban areas, the products showed close agreement with observational estimates. Over rural land defined by the UN Degree of Urbanisation framework, night-time LST increased by $0.41\text{ }^{\circ}\text{C decade}^{-1}$ during 1995–2024. ERA5-Land Tmin showed the same trend magnitude ($0.41\text{ }^{\circ}\text{C decade}^{-1}$), and the NEX-GDDP-CMIP6 ensemble median was also $0.41\text{ }^{\circ}\text{C decade}^{-1}$ (5–95% range: $0.18\text{--}0.63$). Although individual model trends varied, the central estimates reproduced the regional background warming signal, while the additional warming increment observed in urban areas was not represented.

The magnitude of the urban warming increment was compared with projected regional warming from climate scenarios (Fig. S2; Table S2). Relative to the 1995–2014 baseline, NEX-GDDP-CMIP6 projected UAE-area Tmin increases of $0.98\text{--}1.18\text{ }^{\circ}\text{C}$ for the near term (2021–2040), $1.48\text{--}2.45\text{ }^{\circ}\text{C}$ for the mid term (2041–2060), and $1.42\text{--}5.52\text{ }^{\circ}\text{C}$ for the long term (2081–2100) across the four emission scenarios. Over the 2.9 decades between the 1995 and 2024 trend endpoints, the observed Tmin increase at Dubai corresponded to $3.8\text{ }^{\circ}\text{C}$, of which $3.4\text{ }^{\circ}\text{C}$ was not represented by ERA5-Land. The corresponding differences were $2.5\text{ }^{\circ}\text{C}$ at Abu Dhabi and Sharjah and $0.7\text{ }^{\circ}\text{C}$ at Al Ain. At stations with limited urban expansion, including Ras Al Khaimah and Fujairah, observed and product trends remained within their associated uncertainties.

3.5 Projected urban warming to 2050

At the pixel scale, the historical relationship corresponds to an accumulated response of $1.45\text{ }^{\circ}\text{C}$ per 10 pp of built-up expansion (95% CI: $1.21\text{--}1.69$). Averaging expansion within 5 km, the scale used for station comparison and future scenarios, gives $2.37\text{ }^{\circ}\text{C}$ per 10 pp ($1.95\text{--}2.78$). These accumulated, scale-specific responses differ from the trend coefficient of $0.50\text{ }^{\circ}\text{C decade}^{-1}$ per 10 pp.

Retrospective application of the 5 km relationship underestimated the observed station increment by factors of 1.5 at Sharjah, 1.6 at Al Ain, 3.2 at Dubai, and 4.0 at Abu Dhabi. At Ras Al Khaimah and Fujairah, where urban growth was limited, predicted changes of 0.18 and $0.28\text{ }^{\circ}\text{C}$ were comparable in magnitude to observed changes of -0.35 and $0.02\text{ }^{\circ}\text{C}$. The surface-to-air transfer ratio was approximately two across the six stations but was not statistically resolved (Fig. S3b).

UAE built-up surface reached 618 km^2 at the final observation-supported epoch in 2020. Continuing each grid cell's observed 2010–2020 growth rate for 30 years results in $1,068\text{ km}^2$ of built-up surface by 2050 under S-high, compared with 847 km^2 under S-low. Full projection results for every district and growth scenario are given in Supplementary Table S9. The resulting additional night-time surface warming between 2020 and 2050 was estimated at $1.66\text{ }^{\circ}\text{C}$ (95% CI: $1.37\text{--}1.95$) around Sharjah under S-high and $0.86\text{ }^{\circ}\text{C}$ under S-low, 1.18 and $0.59\text{ }^{\circ}\text{C}$ around Abu Dhabi, 1.05 and $0.51\text{ }^{\circ}\text{C}$ around Dubai, 0.77 and $0.38\text{ }^{\circ}\text{C}$ around Al Ain, 0.35 and $0.17\text{ }^{\circ}\text{C}$ at Fujairah, and 0.29 and $0.14\text{ }^{\circ}\text{C}$ at Ras Al Khaimah (Table 4; Fig. 7a).

Alternative growth assumptions produced comparable ranges. The 2005–2020 growth rate resulted in 1.76 and $0.93\text{ }^{\circ}\text{C}$ warming around Sharjah under S-high and S-low, respectively, whereas the 2010–2025 growth rate based on the projected 2025 layer produced 1.14 and $0.58\text{ }^{\circ}\text{C}$. The latter represents the lowest estimate because the projected layer indicates slower post-2020 expansion than observed during the preceding period. For comparison, the NEX-GDDP-

CMIP6 ensemble-median UAE-area Tmin increase for 2041–2060 relative to 1995–2014 was 1.48 °C under SSP1-2.6 (26 models), 1.80 °C under SSP2-4.5 (28 models), 1.96 °C under SSP3-7.0 (23 models), and 2.45 °C under SSP5-8.5 (27 models) (Table S2).

These quantities represent different variables: the projected urban warming increment is a surface-temperature response, whereas NEX-GDDP-CMIP6 projections represent 2 m air temperature. Therefore, the comparison provides a measure of relative magnitude rather than a direct equivalence. Under this framework, the projected urban increment around Sharjah corresponds to 35–112% of the projected mid-century regional Tmin increase depending on scenario and growth pathway, including 92% under SSP2-4.5 with S-high growth and 48% under S-low growth. Under both growth scenarios, the Sharjah increment exceeds the mid-century difference between SSP1-2.6 and SSP3-7.0 projections (0.48 °C). At more slowly growing stations, the relative contribution was smaller; around Dubai, the increment represented 58% under S-high and 28% under S-low of the SSP2-4.5 mid-century warming.

Projection uncertainty was dominated by the conversion from surface temperature to air temperature. The complete variance decomposition and alternative urban-growth scenarios are reported in Supplementary Results R4 and Supplementary Tables S5 and S9.

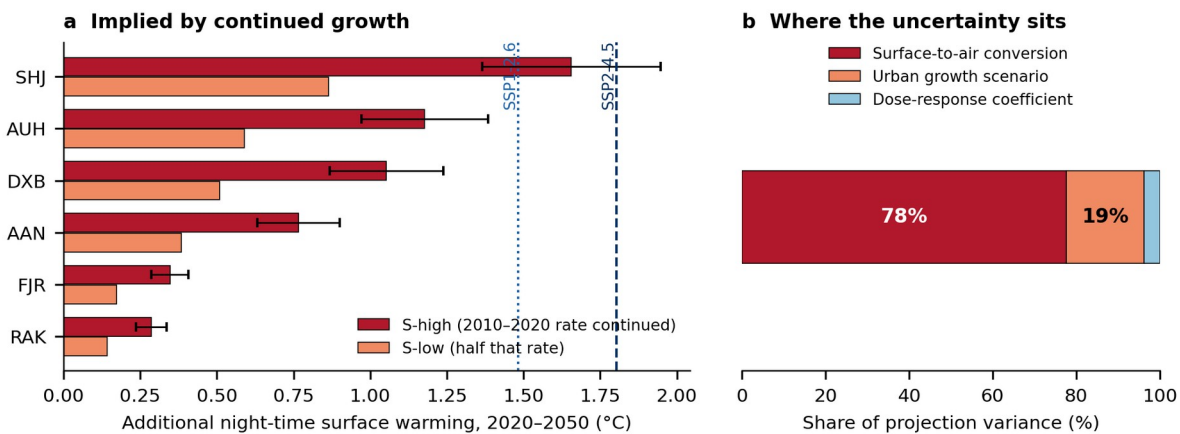


Figure 7. The urban warming increment to 2050 and where its uncertainty lies. (a) Additional night-time surface warming between 2020 and 2050 within 5 km of each station, under continuation of the observed 2010–2020 built-up growth rate (S-high) and half that rate (S-low); error bars are 95% confidence intervals from the exposure–response. Vertical lines give the NEX-GDDP-CMIP6 UAE-area Tmin change for 2041–2060 relative to 1995–2014 under two scenarios; those are 2 m air temperatures and the bars are surface temperatures, so the comparison shows relative magnitude rather than a common variable. (b) Share of the projection variance contributed by the surface-to-air conversion, the built-up growth scenario and the exposure–response coefficient. Station codes as in Fig. 1.

Table 4. Additional night-time surface warming between 2020 and 2050 implied by continued built-up growth within 5 km of each station. S-high continues each cell's observed 2010–2020 growth rate for thirty years and S-low applies half that rate, with both capped at the 99th percentile of observed plan-area fraction. Intervals are 95% confidence intervals from the exposure–response coefficient; full uncertainty is reported in Supplementary Table S5.

Station	Added by 2050 (pp)	S-high (°C)	S-low (°C)
Sharjah	7.0	1.66 (1.37 to 1.95)	0.86
Abu Dhabi	5.0	1.18 (0.97 to 1.38)	0.59

Dubai	4.5	1.05 (0.87 to 1.24)	0.51
Al Ain	3.2	0.77 (0.63 to 0.90)	0.38
Fujairah	1.5	0.35 (0.29 to 0.41)	0.17
Ras Al Khaimah	1.2	0.29 (0.24 to 0.34)	0.14

4. Discussion

4.1 Nocturnal warming mechanisms in hot-desert cities

Four independent results identify an urban warming increment above the regional background. Night-time temperatures diverged from matched controls after land conversion; station-minus-ERA5-Land trends increased with nearby urban expansion; satellite LST exhibited a spatial exposure–response at fixed coastal distance; and stations with little surrounding growth showed little or no increment. These patterns were not reproduced by coastal distance, greening, humidity change, or an isolated station discontinuity. Their agreement across station and satellite records is important because the two data sources have different measurement supports and potential errors. The evidence remains observational and does not establish causation as a controlled experiment would, but the temporal ordering, matched controls, and independent spatial gradients substantially narrow the plausible alternatives.

Surface properties explain part of the association. Adding albedo and night-time light to the spatial model reduced the built-up coefficient by 31–60%, whereas vegetation change produced little attenuation. This is shared statistical explanation rather than causal mediation because built-up fraction, albedo, and light change together during development. The pattern is nevertheless consistent with greater solar-energy storage in darker urban materials and with anthropogenic heat release, including air-conditioning waste heat that can amplify nocturnal temperatures in hot, dry cities (Oke, 1982; Salamanca et al., 2014). The opposite daytime response is also physically plausible: shading, irrigation, and the thermal inertia of urban materials can keep daytime surfaces cooler than bare desert while stored energy is released after sunset (Lazzarini et al., 2015). Stronger summer responses and persistent daytime cooling in urbanising areas without greening support a combination of surface and anthropogenic processes, although direct energy-flux observations are required to partition them.

4.2 Climate-product representation and heat assessment

Previous UAE station–product comparisons could not determine whether large local trends reflected product limitations, station artefacts, or genuine urban change (Ullah et al., 2024). The present results separate these possibilities. ERA5-Land and the NEX-GDDP-CMIP6 ensemble reproduced the rural and regional background but not the positive relationship between urban expansion and nocturnal warming. The discrepancy is consistent with their construction. ERA5-Land runs CHTESSEL offline without a time-varying urban tile, while NEX-GDDP-CMIP6 inherits trends

from global models whose urban land representation is absent or fixed (Muñoz-Sabater et al., 2021; Zhao et al., 2021; Thrasher et al., 2022). ERA5 atmospheric forcing may transmit a weak indirect urban influence, but the near-zero relationship measured between ERA5-Land trends and built-up expansion indicates that this influence is small over the UAE. The persistence of the observed relationship after aggregation to 0.25° further shows that the discrepancy is not solely a resolution effect.

The design also limits a recognised weakness of observation-minus-reanalysis estimates: sensitivity to multidecadal variability in the reanalysis (Wang et al., 2013). The period was fixed by the WMO 30-year convention, stations with little urban growth returned differences indistinguishable from zero, and the independent satellite record produced the same expansion–warming gradient without using reanalysis.

The representation gap matters for urban heat assessment. Products that reproduce regional background warming but omit local urban expansion may understate present night-time temperatures, warm-night frequency, and cooling demand in rapidly growing districts. An observationally constrained urban increment can therefore be evaluated alongside regional projections, provided that LST and air temperature remain distinct and the surface-to-air conversion uncertainty is reported explicitly. The scenario analysis indicates that continued development could make the urban component comparable in magnitude to regional mid-century warming in the fastest-growing districts, but the result is a conditional surface-temperature estimate rather than a correction factor for air-temperature projections. Urban-resolving regional simulations remain the process-based approach required for a predictive estimate.

The same distinction is relevant to operational thresholds. Warm-night indices increased substantially faster at rapidly urbanising stations than in ERA5-Land, including a marked rise in nights above 30 °C at Dubai. Because this component scales with local development rather than greenhouse-gas scenario alone, urban planning can influence part of future nocturnal heat. Assessments should therefore report the regional climate contribution and the locally observed urban increment separately.

4.3 Urban morphology and adaptation implications

The declining marginal response with building height contrasts with the expectation that reduced sky view in taller districts should inhibit nocturnal long-wave cooling (Oke, 1982; Stewart and Oke, 2012). Several explanations remain plausible. Expansion at the urban periphery replaces bare desert, whereas development in established districts more often adds fabric to already modified surfaces. Built-up area may also describe vertical growth poorly, because a small increase in plan area can accompany substantial additional building volume. Shading and thermal inertia could further alter

the balance between daytime heat storage and nocturnal release. The persistence of the interaction after controlling for initial built-up fraction indicates that simple saturation explains only part of the pattern.

These explanations cannot be separated with a single building-height snapshot. Repeated height observations would permit built volume, rather than built area, to be used as the development metric. Land-transition analysis could distinguish desert-to-urban conversion from densification, while sub-daily LST could test whether taller districts exhibit a stronger daytime cool island. The supported inference is narrower: built-up area alone does not fully characterise the thermal implications of development where urban growth is increasingly vertical.

The surface-property analysis identifies candidate adaptation mechanisms but does not quantify intervention effects. Albedo accounted for more shared variation with the urban warming increment than night-time light, while greening was associated mainly with daytime cooling and did not remove nocturnal warming. Reflective materials, vegetation, and urban form may therefore affect different parts of the diurnal cycle. Controlled interventions, direct surface-energy measurements, and urban-resolving simulations are needed before the statistical attenuation estimates can be translated into expected cooling benefits.

4.4 Generalisability, limitations and research priorities

The observed diurnal asymmetry is consistent with evidence from other hot-desert cities. Doha and Phoenix have exhibited stronger long-term warming in T_{min} than T_{max} (Baker et al., 2002; Cheng et al., 2017), while studies of Saudi cities, Kuwait, Abu Dhabi, and Dubai report nocturnal urban warming together with weak or reversed daytime contrasts (Lazzarini et al., 2015; Elhacham and Alpert, 2021; AlKhaled et al., 2024; Munir et al., 2025). The UAE estimates therefore fall within a broader regional pattern rather than representing an isolated thermal regime.

The UAE is particularly informative because rapid, spatially concentrated land conversion provides a strong gradient against an arid rural background. Similar physical conditions occur elsewhere across the Gulf and wider arid subtropics: bare surroundings create a large urban–rural surface contrast, mechanically cooled buildings release anthropogenic heat, and high-thermal-mass materials store daytime energy. These similarities support testing the analytical design elsewhere, but not transferring the UAE coefficients without local calibration.

Because ERA5-Land and NEX-GDDP-CMIP6 are globally applied, urban representation should be evaluated wherever rapid development has altered local thermal conditions. A transferable assessment would combine stations, satellite trends against a harmonised rural reference, and matched comparisons with climate products. Applying this framework across other arid cities would

establish whether the UAE increment is typical, while allowing coefficients to vary with background land cover, building stock, and cooling demand.

Three limitations constrain interpretation. First, satellite LST and 2 m air temperature are distinct quantities; the satellite record establishes spatial structure, while stations estimate the air-temperature response. Their conversion is the largest projection uncertainty (Fig. 7b). Second, the small airport-based station network lacks complete site histories, so the Pettitt test is a homogeneity screen rather than full homogenisation. Station evidence is therefore interpreted jointly with the independent satellite record. Third, building height is available for only 2018, making morphology a cross-sectional modifier rather than a measured change; the LCZ analysis is similarly limited and does not resolve high-rise classes at 1 km.

Additional data limitations affect the mechanism analysis. The satellite series combines several sensors and lacks observations in 2018; urban-minus-rural differencing and sensor-era analyses reduce but do not eliminate this uncertainty. MODIS night-time coverage was too sparse over desert for an independent LST check, Landsat albedo requires narrow-to-broadband conversion, DMSP lights saturate in city centres, and the VIIRS mediator is a 2024 level rather than a temporal change. Night-time light is also only a proxy for anthropogenic heat. ERA5-Land represents the regional background rather than ground truth, and NEX-GDDP-CMIP6 is represented by one realisation per model.

The 2050 calculation is a scenario, not a prediction. It assumes that accumulated warming scales with the magnitude of expansion, that the historical relationship persists as cities densify, and that built-up growth follows the extrapolated GHS trajectories. Priority data needs are therefore a denser urban–rural air-temperature network with documented station histories, repeated building-height observations, and urban-resolving simulations for Gulf cities.

5. Conclusions

Urban expansion was associated with a distinct nocturnal warming increment across the UAE during 1995–2024. Rapidly developing cities warmed substantially faster at night than the regional background, and the increment increased with nearby built-up expansion in both station and satellite records. Matched pixels remained stable before conversion and warmed afterwards, whereas daytime surfaces showed weak warming or cooling. Albedo darkening and night-time light explained part of the spatial association, while the response per unit of added built-up area declined in districts with taller buildings. These results show that the magnitude of urban warming depends on both the extent and form of development.

ERA5-Land and the NEX-GDDP-CMIP6 ensemble reproduced rural background warming but not the additional relationship between urban expansion and night-time temperature. Continued

development could therefore add a locally important component to mid-century warming that is not explicit in regional products. The scenario magnitude remains uncertain because satellite surface temperature cannot be transferred directly to urban air temperature. Urban heat assessments should distinguish this observationally constrained urban increment from regional climate change, retain the associated uncertainty, and avoid treating coarse climate products as complete representations of rapidly changing cities. Denser urban observations, repeated morphology data, and urban-resolving models are required to convert the present empirical relationship into a predictive framework.

Data availability

All primary datasets are publicly available: HadISD (Met Office Hadley Centre), the C3S satellite land-surface temperature record (Copernicus Climate Data Store), GHS-BUILT-S, GHS-BUILT-H, and GHS-SMOD R2023A (European Commission Joint Research Centre), MODIS MOD13A2 and MCD43A3 and Landsat Collection 2 (NASA/USGS), DMSP-OLS and VIIRS night-time lights (NOAA), ERA5-Land (Copernicus Climate Data Store), NEX-GDDP-CMIP6 (NASA Center for Climate Simulation), and GADM 4.1. Processed annual station series, pixel trend fields, urban–rural tables and every derived result reported here are deposited in the public repository at https://github.com/MOHIBULLAH141300/UAE_Urban_Night_Warming, which will be archived with a Zenodo DOI before publication.

Code availability

The complete analysis pipeline — raster stack assembly, pixel trend estimation, the coast-matched exposure–response, station trends and ETCCDI indices, the matched conversion event study, uncertainty decomposition and the projection to 2050 — is available in the same repository (https://github.com/MOHIBULLAH141300/UAE_Urban_Night_Warming). All resampling, bootstrap and permutation procedures use the fixed random seed 20260713, so the reported statistics are exactly reproducible.

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747 The authors declare that they have no known competing financial interests or personal relationships
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